

Problem 1: Grade Evaluation System

Problem Statement: Write a Java program that calculates the average marks of a student and determines the grade based on the following criteria:

Grade A: Average marks ≥ 90

Grade B: Average marks between 70 and 89

Grade C: Average marks between 50 and 69

Grade D: Average marks between 30 and 49

Fail: Average marks < 30

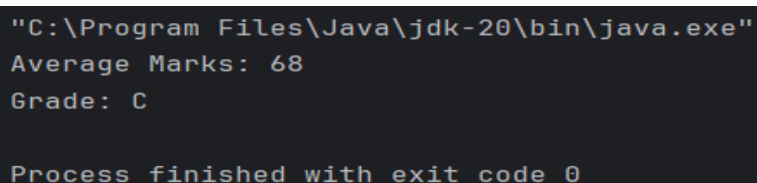
Source Code:

```
package LB_Assingment2;

public class Avg_marks {
    public static void main(String[] args) {
        int Maths=80;
        int Science=50;
        int History=75;

        int avg=(Maths+Science+History)/3;

        if(avg>=90){
            System.out.println("Average Marks: "+avg);
            System.out.println("Grade: A");
        }else if(avg<=89 && avg>=70){
            System.out.println("Average Marks: "+avg);
            System.out.println("Grade: B");
        }else if(avg<=69 && avg>=50){
            System.out.println("Average Marks: "+avg);
            System.out.println("Grade: C");
        }else if(avg<=49 && avg>=30){
            System.out.println("Average Marks: "+avg);
            System.out.println("Grade: D");
        }else{
            System.out.println("Average Marks: "+avg);
            System.out.println("Fail");
        }
    }
}
```

A screenshot of a terminal window showing the execution of a Java program. The command executed is "C:\Program Files\Java\jdk-20\bin\java.exe". The output shows the average marks as 68 and the grade as C. The process finished with exit code 0.

```
"C:\Program Files\Java\jdk-20\bin\java.exe"
Average Marks: 68
Grade: C

Process finished with exit code 0
```

Problem 2: Leap Year

Problem Statement: Write a Java program that checks whether the year is a leap year or not.

Source code:

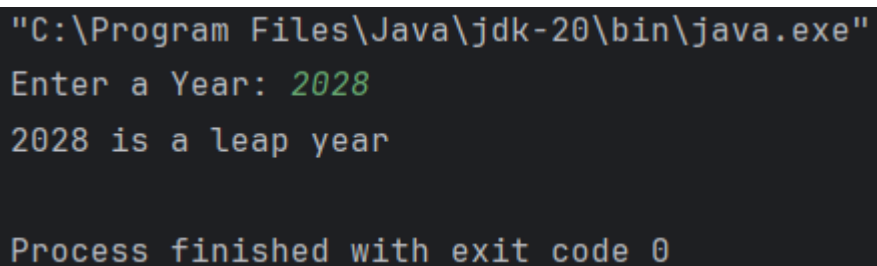
```
package LB_Assingment2;

import java.util.Scanner;

public class Leap_year {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        System.out.print("Enter a Year: ");
        int year=sc.nextInt();

        if((year%4==0 && year%100!=0)|| year%400==0){
            System.out.println(year+" is a leap year");
        }else{
            System.out.println(year+" is not a leap year");
        }
    }
}
```

Output:



```
"C:\Program Files\Java\jdk-20\bin\java.exe"
Enter a Year: 2028
2028 is a leap year

Process finished with exit code 0
```

Problem 3: Days of the Week

Problem Statement: Write a Java program that takes an integer between 1 and 7 and prints the corresponding day of the week using a switch-case statement. If the input is outside the range of 1 to 7, the program should display "Invalid day number".

Source code:

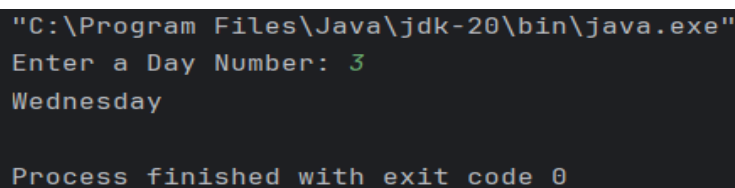
```
package LB_Assingment2;

import java.util.Scanner;

public class DaysOfWeek {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        System.out.print("Enter a Day Number: ");
        int day=sc.nextInt();

        switch (day){
            case 1:
                System.out.println("Monday");
                break;
            case 2:
                System.out.println("Tuesday");
                break;
            case 3:
                System.out.println("Wednesday");
                break;
            case 4:
                System.out.println("Thursday");
                break;
            case 5:
                System.out.println("Friday");
                break;
            case 6:
                System.out.println("Saturday");
                break;
            case 7:
                System.out.println("Sunday");
                break;
            default:
                System.out.println("Invalid day Number");
        }
    }
}
```

Output:



```
"C:\Program Files\Java\jdk-20\bin\java.exe"
Enter a Day Number: 3
Wednesday
Process finished with exit code 0
```

Problem 4: Identify the Values of Uninitialized Variables Scenario: You are working on a program that handles different data types. Your manager has asked you to quickly check the values of various variables, but you're in a rush and forget to initialize them. As you go through the code, you expect some values to show up, but Java has something else in mind. Your task is to fix the issue and ensure the variables hold proper values

Source Code:

```
package LB_Assingment2;

public class Uninitialized_Variables {
    public static void main(String[] args) {
        byte a = 10;
        short b = 1000;
        int c = 100000;
        long d = 100000000000L;
        float e = 10.5f;
        double f = 20.55;
        char g = 'P';
        boolean h = true;

        System.out.println("byte: " + a);
        System.out.println("short: " + b);
        System.out.println("int: " + c);
        System.out.println("long: " + d);
        System.out.println("float: " + e);
        System.out.println("double: " + f);
        System.out.println("char: " + g);
        System.out.println("boolean: " + h);
    }
}
```

Output:

```
"C:\Program Files\Java\jdk-20\bin\java.exe"
byte: 10
short: 1000
int: 100000
long: 100000000000
float: 10.5
double: 20.55
char: P
boolean: true

Process finished with exit code 0
```